

DAVIS ASISH

B.Tech CSE (AI/ML) · Junior AI/ML & Full-Stack Engineer

Chennai, India — voletidavisasish@gmail.com — github.com/davis-asish
— linkedin.com/in/voleti-davis-asish-486995298 — davis-asish.github.io/portfolio

EDUCATION

Bachelor of Technology — Computer Science & Engineering (AI/ML)

SRM Institute of Science and Technology, Kattankulathur, Chennai, India

Expected Graduation: 2027 · Minor in Artificial Intelligence & Machine Learning

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

ML / Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, OpenCV, HuggingFace Transformers, Stable Diffusion

Web & Backend: Next.js 14, React, Flask, Tailwind CSS, Streamlit, Sanity CMS

DevOps & Tools: Docker, Git, GitHub, CUDA, Upstash, Razorpay, Shiprocket

Research Methods: Stratified K-Fold Cross-Validation, SMOTE Oversampling, Gated Multimodal Fusion, CNN + Transformer Architectures

PROJECTS

MMA-Net: Multimodal Arrhythmia Detection

[GitHub](#) — [Live Demo](#)

PyTorch · EfficientNet-B0 · Transformer · Gated Multimodal Unit · MIT-BIH · PhysioNet

- Designed a hybrid CNN + Transformer architecture fusing ECG (electrical) and PCG (acoustic) cardiac signals via a Gated Multimodal Unit (GMU) to classify five AAMI arrhythmia classes.
- Achieved **F1-Macro: 0.855 ± 0.010** , **AUC-ROC: 0.889 ± 0.012** , and **Cohen's Kappa: 0.791 ± 0.011** under stratified 5-fold cross-validation with SMOTE oversampling on MIT-BIH + PhysioNet/CinC 2016.
- Research-grade pipeline with full reproducibility; manuscript under preparation for journal submission.

NeuralStyleTransferApp — Local AI Image Generator

[GitHub](#) — [Live Demo](#)

Python · Flask · PyTorch · Stable Diffusion Turbo (sd-turbo) · Docker

- Built a fully local AI image generation and multi-style transfer web app powered by **stabilityai/sd-turbo** — zero API keys, zero cloud billing.
- Supports text-to-image generation and 6 artistic style presets (Anime, Cyberpunk, Oil Painting, Sketch, Cartoon, Realistic) with auto CUDA/CPU fallback.
- Containerised with Docker for one-command reproducible deployment; images downloadable directly from the browser UI.

Uppucheppa — Premium E-Commerce Platform with AI Chatbot

[GitHub](#) — [Live Demo](#)

Next.js 14 · TypeScript · Tailwind CSS · Sanity CMS · Claude Sonnet · Razorpay · Shiprocket

- Engineered a production-ready e-commerce platform with a full commerce stack: Razorpay payment gateway, Shiprocket logistics integration, Upstash rate limiting, and dynamic Sanity CMS with auto-updating sitemap.
- Integrated **Chitti AI**, a Claude Sonnet-powered streaming chatbot with live product knowledge, Telugu-aware personality, and instant AI knowledge sync from the CMS.
- Implemented *Find My Fish*, a 3-question recommendation quiz matching customers to products by health goal and preference, improving product discoverability.

Password Strength Analyzer

[GitHub](#) — [Live Demo](#)

Python · Streamlit · HaveIBeenPwned API · Docker

- Developed a real-time password security tool with information-theoretic entropy scoring, keyboard-walk and leet-speak pattern detection, and GPU-baseline crack-time estimation.
- Integrated HaveIBeenPwned k-anonymity breach-check API — only 5 SHA-1 prefix characters transmitted, ensuring full user privacy.
- Includes a passphrase and high-entropy random password generator; packaged in Docker for zero-setup deployment.

TicTacToe AI — Unbeatable Game Agent

[GitHub](#) — [Live Demo](#)

Python · PyGame · Minimax · Alpha-Beta Pruning

- Implemented a provably optimal Tic-Tac-Toe AI using Minimax search with Alpha-Beta pruning, guaranteeing perfect play with instant move generation.
- Rendered with PyGame; interactive GUI with colour-coded players and automatic replay on game over.

RESEARCH & ACHIEVEMENTS

Research

MMA-Net: Multimodal Cardiac Arrhythmia Classification via ECG-PCG Fusion

Status: Manuscript under preparation

Proposes a novel Gated Multimodal Unit (GMU) architecture fusing electrical and acoustic cardiac signals. Evaluated on MIT-BIH Arrhythmia Database and PhysioNet/CinC 2016. Achieves state-of-the-art performance on 5-class AAMI arrhythmia classification.

Key Technical Achievements

- Designed and trained a multimodal deep learning system achieving AUC-ROC of 0.889 on a clinically relevant cardiac classification task.
- Shipped a full production e-commerce platform with AI integration, payment gateway, and logistics pipeline as a solo developer.
- Built a privacy-respecting security tool using k-anonymity protocols integrated against a live breach database.
- Delivered 100% local generative AI deployment eliminating cloud cost and latency for end users.

CERTIFICATIONS

GitHub Copilot Certification

Issued: April 25, 2026 Expires: April 26, 2028

Microsoft · GitHub Certification Program

Credential ID: 39510691623FB896 · Certification No: 145F57-RCA45A

[Verify online at Microsoft Learn](#)

SERVICE & PROFESSIONAL DEVELOPMENT

Open Source & Community

- All personal projects published open-source under MIT License on GitHub (github.com/davis-asish) with live demos hosted on GitHub Pages.
- Portfolio website built with React + Vite showcasing projects, skills, and research work.

Professional Development

- Actively deepening expertise in multimodal deep learning, biomedical signal processing, and production AI system design.
- Exploring LLM integration patterns, retrieval-augmented generation (RAG), and agentic AI systems.
- Self-directed study in full-stack TypeScript ecosystem (Next.js, Sanity, Upstash) and cloud-native deployment.